1.6.6 ▶ Driven Oscillations

In damped oscillators, the oscillation eventually dies out, meaning that an external force needs to be supplied to keep the system moving. This force should act with the movement of the object, so that it always is doing positive work on the object. For underdamped oscillators (the only case we will consider, since the other two damping cases don't oscillate), the frequency of this force should thus be the same as the frequency of the damped oscillator, and it should act in phase with the oscillator. Thus, the force must be in the form of

$$F_{driving} = F_0 cos(\boldsymbol{\omega} t) \tag{1.101}$$

where $\boldsymbol{\omega}$ is the natural or **resonant** frequency of the damped oscillator[16], and F_0 is a constant. Adding this driving force to the equation of motion for a damped oscillator yields the following differential equation:

$$m\ddot{x} + b\dot{x} + kx = F_0 cos(\boldsymbol{\omega} t) \tag{1.102}$$

This equation is an **inhomogeneous second order linear ODE**. To solve an inhomogenous ODE, we first find a particular solution to the equation (a couple methods to find one for this particular equation are listed below). Then, we add that particular solution to the solution of the homogeneous ODE[17]. This makes sense, since plugging the solution of the homogeneous equation into the ODE already gives zero, so linearly adding it to another term won't affect anything.

Method 1
Let the particular solution be the function $P(t)$. Plugging this function into eq. 1.91 gives

$$m\ddot{P}(t) + b\dot{P}(t) + kP(t) = F_0 \cos \boldsymbol{\omega} t \tag{1.103}$$

We see that the addition of a combination of different order derivatives of $P(t)$ yields a cosine function. Logically, then, $P(t)$ should be a sine or a cosine function, since they're each other's derivatives, and any sin or cosine function can be represented as a sum of other sine and cosine functions. Thus, we can confidently guess that the particular solution is of the form

$$P(t) = A \cos(\boldsymbol{\omega} t + \phi)$$

Where A and ϕ are constants we will solve for later.

Method 2
Let the particular solution be the function $P(t)$. Using the reasoning discussed above, the most general solution to the differential equation will look like

$$x(t) = A_0 e^{-t/\tau} \cos(\boldsymbol{\omega}' t + \phi_0) + P(t).$$

As time passes, the first term will decay to zero, leaving just the second term:

$$x(t)|_{t\to\infty} = P(t)$$

Now, we examine the purpose of the driving force in the first place. We added this force for the sole purpose of keeping the oscillator oscillating after a long time. Thus, we should

[16] We labeled this quantity as $\boldsymbol{\omega}'$ in 1.6.4
[17] i.e. setting the right hand side of eq. 1.91 to be zero.

expect that with this driving force, the oscillator continues to oscillate as time goes to infinity. Hence, the particular solution must just be the equation of motion of a simple harmonic oscillator i.e.

$$P(t) = A\cos(\omega t + \phi)$$

where A and ϕ are constants we will solve for later.

Now, we solve for A and ϕ. Plugging our particular solution into the differential yields

$$-m\omega^2 A\cos(\omega t + \phi) - b\omega A\sin(\omega t + \phi) + kA\cos(\omega t + \phi) = F_0\cos\omega t.$$

Expanding using additive trig identities yields the following massive equation that doesn't fit in a single line

$$-m\omega^2 A(\cos\omega t\cos\phi - \sin\omega t\sin\phi) - b\omega A(\sin\omega t\cos\phi + \cos\omega t\sin\phi)$$
$$+kA(\cos\omega t\cos\phi - \sin\omega t\sin\phi) = F_0\cos\omega t.$$

Factoring out the $\sin\omega t$ and $\cos\omega t$ terms yields

$$((kA - m\omega^2 A)\cos\phi - b\omega A\sin\phi)\cos\omega t$$
$$+((-kA + m\omega^2 A)\sin\phi - b\omega A\cos\phi)\sin\omega t = F_0\cos\omega t.$$

Since we know that the coefficients of the $\cos\omega t$ terms add up to F_0 and the coefficients of the $\sin\omega t$ terms add up to 0, we have the following system of equations

$$(kA - m\omega^2 A)\cos\phi - b\omega A\sin\phi = F_0 \tag{1.104}$$

$$(-kA + m\omega^2 A)\sin\phi - b\omega A\cos\phi = 0 \tag{1.105}$$

The full steps to solve this system will not be included here, since they are very messy. However, an overview will be given and the reader is encouraged to follow the general steps to solve the system. Solving for ϕ is easiest, and is simply done by rearranging the second equation to yield

$$\tan\phi = \frac{b\omega}{k - m\omega^2}. \tag{1.106}$$

Finding A requires rearranging the first equation and cleverly using the pythagorean identity $\tan^2\theta + 1 = \sec^2\theta$ to write everything in terms of $\tan\phi$. Then, plugging in $\tan\phi$ and reducing yields the result

$$A = \frac{F_0}{\sqrt{(k - m\omega^2)^2 - b^2\omega^2}} \tag{1.107}$$

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